



Day 13 : Storage and Virtual Machines

Cloud Computing and Big Data Analytics — Digital Science Programme

Agenda — Today's Journey

- **Lesson 1: Cloud Storage Fundamentals — Block, Object, and File Storage**
- **Lesson 2: Block Storage — Virtual Disks in the Cloud**
- **Lesson 3: Object Storage — The Data Lake Backbone**
- **Lesson 4: File Storage and Network-Attached Storage (NAS)**
- **Lesson 5: Introduction to Virtual Machines (VMs) and Hypervisors**
- **Lesson 6: VM Lifecycle Management — Provisioning, Scaling, and Termination**
- **Lesson 7: Storage and VM Integration — Attaching and Managing Volumes**
- **Lesson 8: Snapshots, Backups, and Disaster Recovery (DR)**

LESSON 1

Cloud Storage Fundamentals Block, Object, and File Storage

Block Storage (1/3)

- Data is split into fixed-size chunks called blocks, each with a numeric address
 - No filenames, no metadata — just raw bytes and addresses
- Fastest access speed, highest cost per GB, typically single-attach to one VM
 - Think of it as numbered storage lockers in a warehouse
- **Cloud services:**
 - AWS → Amazon EBS (Elastic Block Store)
 - Azure → Azure Managed Disks
 - GCP → Google Persistent Disk
- Best for: databases, boot volumes, high-performance applications

Object Storage (2/3)

- Data stored as self-describing objects: DATA + METADATA + UNIQUE ID
 - Flat namespace — no real directories (slashes are part of the key string)
- Cheapest at scale, near-infinite scalability, higher latency than block
 - Immutable — cannot edit a byte in place; must re-upload the entire object
- **Cloud services:**
 - AWS → Amazon S3 (Simple Storage Service) — 350+ trillion objects
 - Azure → Azure Blob Storage
 - GCP → Google Cloud Storage (GCS)
- Best for: data lakes, ML training data, backups, media archives

File Storage (3/3)

- Traditional folder hierarchy served over the network via NFS or SMB protocols
 - NFS (Network File System) — dominant on Linux
 - SMB (Server Message Block) / CIFS — dominant on Windows
- Killer feature: SHARED ACCESS — multiple VMs mount the same filesystem
- **Cloud services:**
 - AWS → Amazon EFS (Elastic File System) — auto-scaling NFS
 - Azure → Azure Files — supports SMB + NFS
 - GCP → Google Filestore — high-performance NFS
- Best for: shared datasets, CMS, distributed ML training

Storage Paradigm Comparison

Feature	Block	Object	File
Structure	Fixed-size chunks	Flat namespace	Hierarchical
Access	Device (iSCSI/NVMe)	API (HTTP/S)	Protocol (NFS/SMB)
Metadata	Minimal	Rich & custom	Standard
Latency	Sub-millisecond	Milliseconds–seconds	Medium
Scalability	Limited	Near-infinite	Moderate
Attach Model	Single VM	Stateless API	Multiple VMs
Best For	Databases, OS	Data lakes, backup	Shared datasets



KEY TAKEAWAY

As a Data Scientist, object storage is your primary workhorse.
It backs every data lake, ML training pipeline, and analytics warehouse in the cloud.
Block is for speed. File is for sharing. Object is for scale.

LESSON 2

Block Storage Virtual Disks in the Cloud

Block Storage Deep Dive

- A network-attached virtual disk that behaves like a physical SSD or HDD
 - Created via API in seconds, formatted (ext4/NTFS) and mounted inside a VM
- Every VM needs at least one: the boot volume that holds the OS
- **Performance is measured by two metrics:**
 - IOPS — Input/Output Operations Per Second (read/write speed)
 - Throughput — MB/s (data transfer rate)
- Key constraint: typically single-attach (one volume → one VM)
 - Must reside in the same Availability Zone (AZ) as the VM

AWS EBS Volume Types

Type	Max IOPS	Max Throughput	Best For
gp3 (GP SSD)	16,000	1,000 MB/s	Most workloads
io2 Block Express	256,000	4,000 MB/s	Critical databases
st1 (Throughput HDD)	500	500 MB/s	Big data streaming
sc1 (Cold HDD)	250	250 MB/s	Infrequent archival

EBS Pricing Model (gp3 Example)

- **Region: us-east-1 (N. Virginia)**
- Storage: \$0.08 per GB-month
- Baseline IOPS: 3,000 included free with every gp3 volume
- Additional IOPS: \$0.005 per provisioned IOPS-month
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- **Example Calculation:**
 - 200 GB volume $\times$ \$0.08 = \$16.00 (storage)
 - 6,000 IOPS – 3,000 baseline = 3,000 extra $\times$ \$0.005 = \$15.00
 - **Monthly Total = \$31.00**



KEY TAKEAWAY

Block storage is where your databases, Jupyter servers, and operating systems live.

Master IOPS and throughput metrics to avoid performance bottlenecks.

A 200 GB high-performance volume costs only ~\$31/month.

LESSON 3

Object Storage The Data Lake Backbone

Anatomy of an Object

- **Every object contains three components:**
- 1. DATA — the binary payload (up to 5 TB per object in S3)
- 2. METADATA — system-generated + custom key-value pairs
 - Examples: project name, data owner, row count, sensitivity classification
- 3. UNIQUE KEY — the object's address within a bucket / container
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- **The Flat Namespace Illusion:**
- `s3://my-bucket/data/2026/april/sales.csv`
 - The forward slashes are part of the key string — NOT real directories
 - Console renders them as folders for human convenience only

Object Storage Tiers (S3 Example)

Tier	Access Pattern	Retrieval Time	Relative Cost
S3 Standard	Frequent (daily)	Milliseconds	\$\$\$\$
S3 Infrequent Access	Monthly	Milliseconds	\$\$\$
S3 Glacier Instant	Quarterly	Milliseconds	\$\$
S3 Glacier Deep Archive	Yearly / never	12–48 hours	\$

The WORM Pattern & Data Science

- **WORM = Write Once, Read Many**
 - Data is written once and read thousands of times by analytics engines
 - Cannot edit byte 47 in place — must re-upload the entire object
- Perfect for Spark, Pandas, TensorFlow pipelines
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- **Why Data Scientists care:**
 - Data lakes are built entirely on object storage
 - ML training datasets (images, text corpora, CSVs) live here
 - Model artefacts (weights, checkpoints) are stored here
 - Versioning enables experiment reproducibility



KEY TAKEAWAY

Object storage is cheap, infinitely scalable, and self-describing. Understanding storage tiers alone can save your organisation thousands of dollars per year on cloud bills.

LESSON 4

File Storage and Network-Attached Storage (NAS)

File Storage in the Cloud

- **What makes it special:**
- **SHARED ACCESS** — dozens to thousands of VMs mount the same filesystem
 - Traditional hierarchy: directories, subdirectories, files
 - Protocols: NFS (Network File System) for Linux, SMB for Windows
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- **Cloud services:**
 - AWS → Amazon EFS — auto-grows/shrinks, NFS v4.1
 - Azure → Azure Files — supports both SMB and NFS
 - GCP → Google Filestore — high-performance NFS
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- You pay only for storage consumed — no pre-provisioning needed

Use Case: Distributed ML Training

- **Scenario:**
- 3 GPU worker nodes training a computer vision model simultaneously
 - All 3 mount the SAME EFS share at /mnt/shared-data/
 - Shared labels.csv + partitioned image batches per worker
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- **Trade-offs:**
-  Familiar — just like a local drive
-  Multi-attach — many VMs at once
-  Costlier per GB than object storage
-  Slower than block storage



KEY TAKEAWAY

When multiple compute nodes need simultaneous access to the same dataset, file storage is the answer. It is the glue for distributed training jobs and shared configuration files.

LESSON 5

Virtual Machines and Hypervisors

What Is a Virtual Machine?

- A software-defined computer — CPU, RAM, storage, NIC — all emulated
 - Runs its own OS, completely isolated from other VMs on the same host
- Created, started, stopped, and destroyed in seconds
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- **The Hypervisor:**
- Software layer between physical hardware and VMs
 - Carves up host resources and allocates them in isolation
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- **Type 1 (Bare-metal) — runs directly on hardware**
 - KVM, VMware ESXi, Microsoft Hyper-V — used in cloud data centres
- **Type 2 (Hosted) — runs on top of a host OS**
 - VirtualBox, VMware Workstation — used for dev/test labs

Multi-Tenancy — The Cloud Business Model

- **One physical server (\$15,000) can host 20–100 VMs**
 - Each VM runs a different customer's workload — complete isolation
 - This is why cloud computing is cheaper than owning hardware
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- **Example: 64-core host with 512 GB RAM and 4 TB SSD**
 - VM-1 Jupyter Hub: 8 vCPU | 64 GB RAM | 500 GB
 - VM-2 PostgreSQL DB: 16 vCPU | 128 GB RAM | 1 TB
 - VM-3 Web Server: 4 vCPU | 16 GB RAM | 100 GB
 - VM-4 CI/CD Runner: 8 vCPU | 32 GB RAM | 200 GB
 - **Remaining: 28 vCPU | 272 GB RAM | 2.2 TB**



KEY TAKEAWAY

Every cloud instance you launch is a VM running on a hypervisor.
Understanding resource allocation and multi-tenancy
is the foundation of cloud architecture and cost optimisation.

LESSON 6

VM Lifecycle Management Provisioning, Scaling, and Termination

The VM Lifecycle States

- **PENDING → RUNNING → STOPPED → TERMINATED**
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- Provision — Choose AMI/image, instance type, VPC, security group
 - Start — VM boots, begins billing immediately
- Stop — VM pauses; storage persists; most billing stops
 - Resize — Must stop first → change instance type → restart
- Terminate — VM destroyed; all ephemeral resources released
 - ⚠ You cannot restart a terminated instance

Scaling Strategies

- **Vertical Scaling (Scale UP):**
- Make the VM bigger — more CPU, more RAM
 - Requires a reboot; used for monolithic applications
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- **Horizontal Scaling (Scale OUT):**
- Add MORE VMs behind a load balancer
 - No downtime; used for distributed applications
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- **Common Beginner Mistake:**
- Forgetting to TERMINATE lab instances
 - A running p3.2xlarge (GPU) = ~\$3.06/hr = \$73/day = \$2,200/month!

Provider Terminology Cheat Sheet

Concept	AWS	Azure	GCP
VM Service	EC2	Virtual Machines	Compute Engine
OS Template	AMI	Image	Machine Image
Firewall	Security Group	NSG	Firewall Rule
Network	VPC	VNet	VPC
Block Storage	EBS	Managed Disks	Persistent Disk
Object Storage	S3	Blob Storage	Cloud Storage
File Storage	EFS	Azure Files	Filestore



KEY TAKEAWAY

Every minute a VM runs costs money. Every minute it sits idle wastes money.
Master the lifecycle — especially TERMINATION —
to avoid surprise cloud bills.

LESSON 7

Storage and VM Integration Attaching and Managing Volumes

How Storage Connects to VMs

- **Block Storage — attach, format, mount**
 - Like plugging a USB drive into your computer (but over the network)
 - Must be in the same Availability Zone as the VM
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- **Object Storage — access via HTTP API**
 - No "attach" operation — stateless REST calls from anywhere
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- **File Storage — mount via NFS/SMB**
 - Like mapping a network drive — appears as a local directory

The Attach-Format-Mount Workflow

- Step 1: ATTACH — Create EBS volume, attach to EC2 via console or CLI
 - Volume appears as `/dev/xvdf` inside the Linux VM
- Step 2: FORMAT — `sudo mkfs.ext4 /dev/xvdf`
 - Creates ext4 filesystem on the raw block device
- Step 3: MOUNT — `sudo mount /dev/xvdf /mnt/data`
 - Makes the volume accessible at `/mnt/data`
- Step 4: PERSIST — Add entry to `/etc/fstab`
 - Without this, the mount is lost on reboot!

Best Practice: Separate OS and Data

- **Root volume (/dev/xvda) — Operating system ONLY**
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- **Data volume (/dev/xvdf) — Databases, notebooks, logs**
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- **Why this matters:**
- Terminate and rebuild the VM without losing data
 - Detach data volume → attach to a new VM in minutes
 - Snapshot data volumes independently for backups
 - Upgrade OS without touching application data



KEY TAKEAWAY

Always separate your OS from your data.
This single practice prevents data loss,
simplifies upgrades, and enables clean disaster recovery.

LESSON 8

Snapshots, Backups, and Disaster Recovery (DR)

What Is a Snapshot?

- **A point-in-time copy of a block storage volume**
- First snapshot = full copy of all data blocks
 - Subsequent snapshots = INCREMENTAL (only changed blocks)
- Stored in object storage behind the scenes (S3 for AWS)
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- **Snapshot Superpowers:**
 1. Backup & Restore — create new volume from snapshot in minutes
 2. Cross-Region Replication — copy to another region for DR
 3. VM Cloning — AMI = snapshot + boot config = golden image

Incremental Snapshot Savings

- **Scenario: 500 GB PostgreSQL volume, ~10 GB changes/day**
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- **Full Backup Strategy:**
- $500 \text{ GB} \times 30 \text{ days} = 15,000 \text{ GB} \rightarrow \$750/\text{month} @ \$0.05/\text{GB}$
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- **Incremental Snapshot Strategy:**
- Day 1: 500 GB (full) + Days 2–30: 290 GB (deltas) = 790 GB $\rightarrow \$39.50/\text{month}$
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- **SAVINGS: 14,210 GB saved $\rightarrow \$710.50/\text{month} \rightarrow 94.7\%$ reduction**

DR Metrics and Automation

- **Two critical metrics:**
- RPO (Recovery Point Objective) — max acceptable data loss
 - Hourly snapshots → 1-hour RPO
- RTO (Recovery Time Objective) — max acceptable downtime
 - Restore from snapshot → ~15 minute RTO
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- **Automation Services:**
- AWS → Data Lifecycle Manager (DLM)
- Azure → Azure Backup
- GCP → Scheduled Snapshots
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- Never rely on manual backups in production!



KEY TAKEAWAY

Incremental snapshots save 90%+ on backup costs compared to full copies.
Automate your snapshot schedule — set your RPO/RTO targets
and let the cloud handle the rest.

Summary — Eight Lessons at a Glance

#	Lesson	Core Concept
1	Cloud Storage Fundamentals	Block (fast) · Object (scale) · File (share)
2	Block Storage	Virtual disks; IOPS + throughput metrics
3	Object Storage	Flat namespace; data lake backbone
4	File Storage / NAS	Multi-attach; NFS / SMB protocols
5	VMs & Hypervisors	Software-defined compute; multi-tenancy
6	VM Lifecycle	Provision → Run → Stop → Resize → Terminate
7	Storage + VM Integration	Attach, format, mount; separate OS from data
8	Snapshots & DR	Incremental saves 95%; automate RPO/RTO

What's Next?

- **Lab Exercise:**
 - Launch an EC2 instance, attach an EBS volume, format and mount it
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- **Homework:**
 - Calculate monthly costs for a 3-VM architecture using the AWS Pricing Calculator
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- **Next Session:**
 - Containers, Docker, and the shift from VMs to microservices



Thank You!

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